



KARAN RAJ

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[LINKEDIN](#) | [GITHUB](#) | [LEETCODE](#)

EDUCATION

Course	College/School	Year	CGPA/Percentage
B.Tech (ECE)	Netaji Subhas University of Technology	2023-2027	8.02
CBSE Board (Class XII)	Bal Bhavan Public School	2023	88%
CBSE Board (Class X)	Bal Bhavan Public School	2021	96%

PROJECTS

Smart RAG System – Multimodal RAG Backend (FastAPI, FAISS, Groq)

- Built a **modular Retrieval-Augmented Generation backend** supporting **PDF, DOCX, image, audio, URL and YouTube parsing**.
- Integrated **E5 embeddings**, reranking, **ChromaDB memory**, **LLaMA3 via Groq**, **web search fallback**, **TTS and file conversion**.

Built a GPT-style transformer from scratch in PyTorch

- Implementing **tokenization**, **positional encoding**, masked self-attention, and **decoder-only architecture**.
- Trained it to generate sentence completions and deployed it as a **basic interactive chatbot**.

Stock Forecasting Web Application

- Built a **Streamlit web app** for predicting stock prices using **XGBoost** and live Yahoo Finance data.
- Integrated model loading, feature scaling, and **visual feedback in a real-time browser UI** which displays 60-days forecasts with matplotlib charts.

EXPERIENCE & LEADERSHIP

App Dev Member at Google Developers Student Club (GDSC)

Played a key role in organizing bootcamps at GDSC, managing logistics and participant engagement.
Participated in hackathons organized by GDSC.

Event Manager At Moksha NSUT

Contributed to the organization of the 2024 Moksha cultural fest at NSUT, assisting with event planning and Execution.

ACHIEVEMENTS & CERTIFICATIONS

Secured an AIR under 50 in the IMUCET examinations (Merchant Navy).
JEE ADVANCED 2023 QUALIFIER.
Secured 98 percentile in JEE MAINS 2023
Solved 200+ problems across LeetCode, GFG.
Represented School in various zonal level olympiads.

SKILLS

Programming Languages: C++, Python,Kotlin(basic)

Tech Stack: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn ,Streamlit,LLM,Kotlin

Proficiency in: Data Structures and Algorithm ,STL

Languages: English, Hindi, French(beginner)